Early Prediction of Sepsis

***Abstract*—Early prediction of sepsis is clinically important for supporting rapid recognition and treatment, which is essential for improving patient outcomes. However, this remains challenging due to early symptoms often being non-specific. This study aimed to implement a range of machine learning models, specifically Logistic Regression, XGBoost and GRU variations to investigate early prediction and extract useful insights for clinicians.**

The study uses the PhysioNet 2019 Challenge dataset, which contains hourly physiological, laboratory and demographic measurements. Several pre-processing and feature engineering techniques are explored, including imputation, missingness indicators, and temporal trend-based features. Post-processing is also explored in terms of feature ablation and threshold tuning effects.

Overall, the model achieved an F1 of 0.198 and a utility of 0.293 in comparison to the highest-scoring challenge paper which achieved a 0.36 utility. The results confirm wider medical literature on sepsis indicators and also highlight the effects of hospital monitoring and recording practices on the dataset and model generalisability across hospitals. Interpretable insights are developed based on SHAP analysis of model performance and clustering results in the categorisation of two distinct pre-septic patient groups based on treatment.

I. Introduction

Sepsis is a life-threatening condition in which the body produces a systemic response to infection that, if not treated early enough, can progress to organ failure and death [1]. It is one of the most common causes of death in intensive care units [2], accounting for 9% of all deaths in the US.

Rapid recognition and treatment are crucial for improving patient outcomes; however, this remains challenging due to a lack of consistency in clinical assessment [3]. Early symptoms are often non-specific and can resemble other forms of clinical deterioration, making sepsis difficult to identify before a patient becomes critically unwell [3].

Machine learning offers a potential approach for supporting earlier prediction of sepsis by identifying patterns in routinely collected clinical data. Hospital patients are regularly monitored through measurements, such as vital signs and laboratory results, which may contain early indicators of deterioration before a clinical diagnosis can be made [4]. By learning from these patterns, predictive models could help identify patients at increased risk and provide additional information to support clinical decision-making.

However, applying machine learning to early sepsis prediction presents several challenges. Clinical data is often noisy, incomplete and recorded at irregular intervals, as measurements are usually taken according to clinical need rather than at fixed time points. In addition, sepsis prediction is typically an imbalanced classification problem, as only a small proportion of patients develop sepsis compared with those who do not. It is also important that predictions are made using only information available before sepsis onset, to avoid data leakage and ensure that the model reflects realistic early prediction. These challenges are considered throughout this study.

Therefore, this study aims to develop and evaluate machine learning models to support the prediction of sepsis up to six hours before the clinical diagnosis of sepsis, using collected ICU data. Different modelling approaches are compared to assess the feasibility of early prediction and identify which clinical features are most informative. In practice, these models could support clinical decision-making by flagging patients who may require closer monitoring or earlier review, rather than replacing clinical judgement.

A. Background and Related Work

Machine learning has been explored as a method of improving sepsis prediction using routinely collected electronic health record data. Henry et al. developed the TREWScore model to predict septic shock from physiological and laboratory data, reporting an AUROC of 0.83 and identifying patients a median of 28.2 hours before onset [5]. More recently, systematic reviews have suggested that machine learning models for sepsis prediction often achieve promising performance, but results vary substantially due to differences in datasets, sepsis definitions, missing data handling and evaluation methods [6].

This study is based on the 2019 PhysioNet challenge, which focused on predicting sepsis from routinely collected clinical data [7]. As part of the challenge, models are evaluated using a utility score designed specifically for early sepsis prediction. Rather than only measuring whether predictions are correct or incorrect, the utility score also considers when a prediction is made. Early correct predictions are rewarded because they are more likely to allow time for clinical intervention, while missed sepsis cases, late predictions and false alarms are penalised. This score is therefore more appropriate for evaluating early-warning models, where the clinical usefulness of a prediction is strongly affected by its timing.

Several submissions to the challenge used tree-based ensemble methods. For example, Zabihi et al. used an ensemble of XGBoost models with features derived from both observed and missing clinical data, achieving third place with an official utility score of 0.339 [8]. Sarafrazi et al. also found that XGBoost outperformed sequential models, suggesting that engineered features and missing-data handling can have a large impact on the final performance [9].

TABLE I
SUMMARY STATISTICS FOR THE DATASET

Metric	Hospital A	Hospital B	Combined
No. of patients	20,336	20,000	40,336
No. of septic patients	1,790	1,142	2,932
% of patients with sepsis	8.80	5.71	7.27
No. of rows with sepsis	17,136	10,780	27,916
% of rows with sepsis	2.17	1.41	1.80
No. of entries	5,536,849	4,950,064	10,486,613

II. The Dataset

The dataset, sourced from the PhysioNet challenge, contains a decade of patient data from two geographically distinct U.S. hospitals: Beth Israel Medical Center in Detroit (hospital A) and Emory University Hospital in Atlanta (hospital B). Across the two hospitals, hourly data is available for 40,336 patients, including demographic information, 8 vital signs and 40 lab measurements. At each time step a binary sepsis label indicates if the patient is septic or diagnosed as septic in the following 6 hours. The summary table for the dataset can be seen in Table I.

The summary statistics highlight the large class imbalance in the dataset, with only 7.27% of patients in the dataset becoming septic. Moreover, as only six to ten hours of septic data are stored for each septic patient, only 1.80% of the rows in the dataset are actually marked as positive.

A. Data preparation

For each hospital, patient data is structured as individual files containing hourly time-series measurements spanning the duration of each hospital stay. All patient files are concatenated to form a single hospital-level dataset. A binary indicator is then assigned to each patient to indicate the patient's hospital.

B. Privacy and Ethics

As a publicly available medical dataset containing confidential patient records, the dataset is already highly anonymised with any directly identifiable information previously removed.

However, two quasi-identifiers remain in the dataset: age and gender. Given that the locations of the hospitals are known, there is a slight risk of a background knowledge attack, whereby someone with access to another dataset or additional information may be able to re-identify patients. To mitigate this privacy risk, ages are recoded into categorical variables with bins of width 10. This generalisation increases the k-anonymity from 1 to 9, meaning that, for each hospital, there are at least 9 patients who share age bins, gender and sepsis outcome. To ensure there isn't a risk of a homogeneity attack, it is also necessary to satisfy l-diversity. After recoding age bins, the l-diversity also increases from 1 to 2, meaning that all sepsis outcomes, a positive or negative diagnosis, are present for each grouping. This prevents a diagnosis from being disclosed when a grouping shares identical quasi-identifiers.

Ethically, the primary issue to be aware of is any potential gender bias introduced in predictive modelling. Biological differences across genders can influence the susceptibility and progression of sepsis, and therefore, it is essential for any models to learn patterns for both groups. However, as males are slightly overrepresented in the number of patients and exhibit a higher prevalence of sepsis, there is a risk that the models bias the majority group. To address this, model performance is evaluated across genders and, where necessary, measures will be taken to ensure fairness in predictive outcomes.

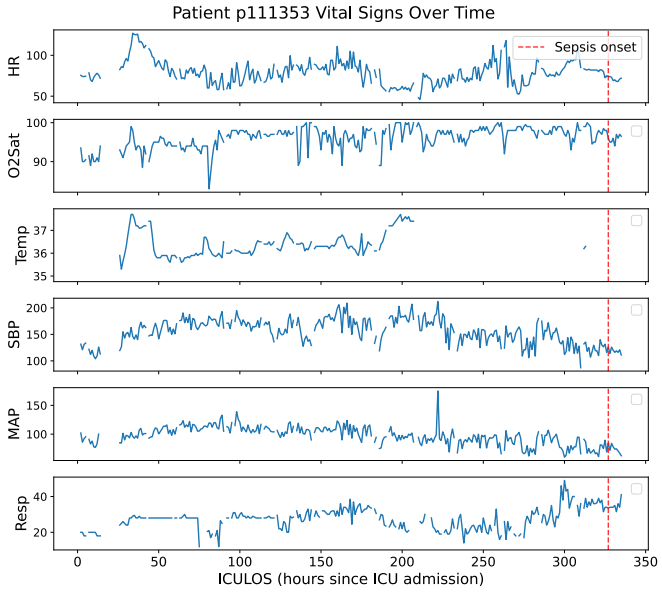


Fig. 1. Example time series for patient data (patient p111353), the sepsis onset point, 6 hours before diagnosis is marked in red.

III. Data Exploration

An example time series displaying a subset of patient vitals signs is shown in Figure 1. Investigating the early sepsis onset point, six hours before diagnosis, there is very little to indicate a patient is developing sepsis. This underscores the difficulty of accurate early sepsis detection and reinforces the need for machine learning to help capture underlying patterns to support doctors in diagnosis. The time series also highlights a clear challenge found in medical datasets: the large number of missing data points.

The distributions of patient vital signs across septic and non-septic points is shown in Figure 2.

Investigating the overall distributions of the features, large skews are exhibited across multiple features, including O2Sat, MAP, DPB and Respiration rate. The high number of outliers across the distributions is also notable. Although outlier removal is considered, extreme values are common in ICU data and often reflect true medical states; as such, removing them would risk the loss of clinically relevant information. Comparing the distributions of septic and non-septic data points, they are very similar across all vital signs. However, patients undergoing sepsis onset exhibit slightly higher heart rates, temperature and respiration rates, alongside slightly

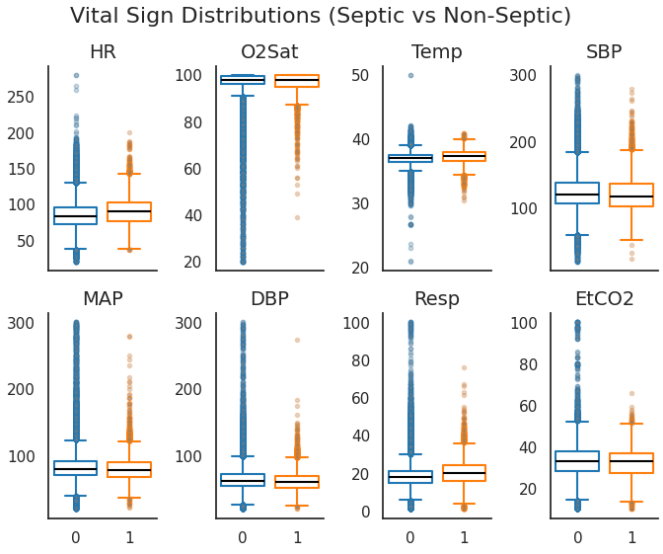


Fig. 2. Box plots showing the distribution of vital signs for non-septic (0) and septic patients (1). The white line represents the median of the distribution and outliers.

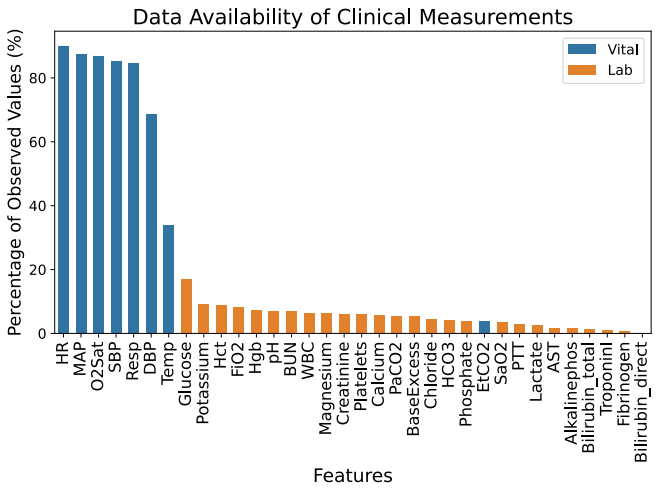


Fig. 3. Proportions of available data for each feature in the dataset

lower DBP, SBP and MAP, consistent with the expected medical behaviour seen in septic patients.

A. *Missingness and Imputation*

As highlighted in Figure 1, medical datasets are characterised by large amounts of missing data, with measurements often not recorded at every hour of an ICU stay. The percentage of observed variables for each clinical measure is shown in Figure 3. Vital signs are generally well represented in the dataset, with only temperature and EtCO2 having less than 60% of data recorded. EtCO2 is also unique in the fact that it is the only measurement not represented in both datasets, only being measured in hospital B. In comparison, lab data points are more sparsely populated, with measurements often only taken when requested by a doctor instead of at regular intervals.

The magnitude of missing data presents significant challenges for data analysis and predictive modelling, and as such, different imputation techniques are evaluated for their accuracy when filling time-series data.

Forward Filling (FF) is the simplest method, propagating the last recorded measurement until the next known value [10]. Linear interpolation (LI) is an extension where missing values are filled along a straight line between known values. Multivariate imputation (MI) is implemented to capture non-linear relationships between related features and temporal patterns across patients [11]. After each method, median filling is implemented to fill any remaining missing values.

To test the effectiveness of the strategies for different levels of missing data, each is tested on varied proportions of masked data points. The RMSE between each imputed dataset and the original is used to assess the effectiveness of each method, with the results shown in Figure 4.

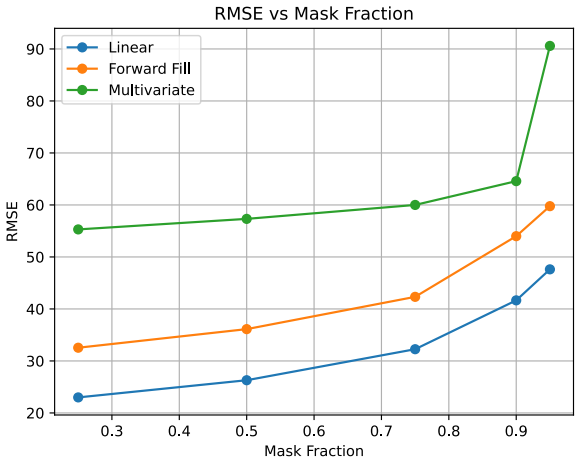


Fig. 4. Comparing the RMSE for the preprocessing methods when trying to impute different levels of missing data.

LI imputed the data with the least error across all masking fractions. FF gave marginally greater error than LI but scaled very similarly when more missing data was introduced. On the contrary, MI produced the highest imputation error and exhibited the worst scaling behaviour. These results inform imputations later used in t-SNE analysis and machine learning.

B. *Demographic analysis*

Demographic analysis is performed to asses how sepsis diagnosis rates vary across different gender and age groups for patients in the ICU. The results are shown in Figure 5.

Across most age ranges men are exhibit higher sepsis rates, with an overall rate of 7.70% compared with the female value of 6.71%. This is expected due to differing immune responses, men are biologically more vulnerable to sepsis onset. The distribution of sepsis across different ages shows sepsis rates do not vary significantly, except for the females between 10

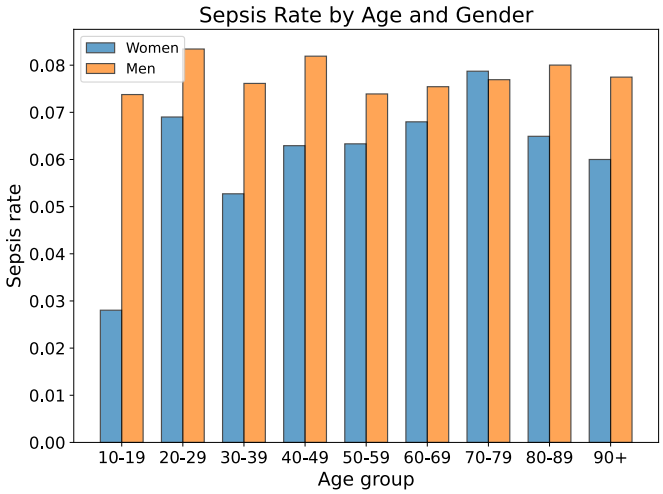


Fig. 5. Demographic analysis comparing how age and gender impact the rate of sepsis.

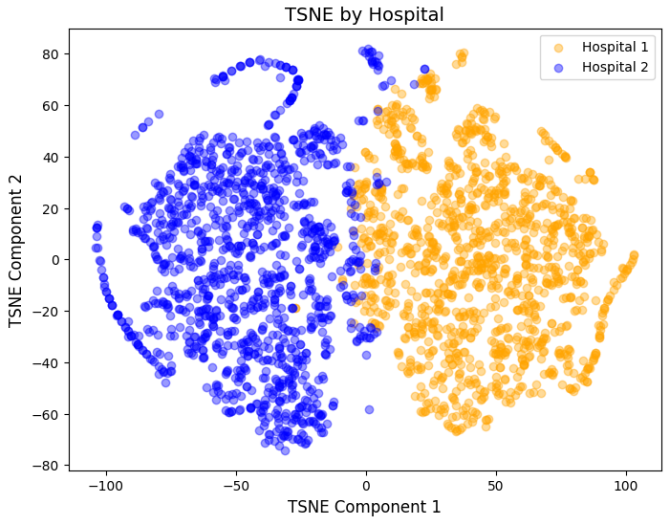


Fig. 6. TSNE dimensionality reduction showing separability across the two hospitals

and 19. However, this difference may arise from the small sample size for this demographic group.

C. Dimensionality Reduction

To understand how the data is clustered across the high-dimensional feature space, t-SNE (t-distributed Stochastic Neighbour Embedding) analysis is performed. T-SNE is not compatible with missing values, and as a result, the dataset is imputed using LI, the most accurate method from section III-A. Before analysis is performed, features are normalised and any skewed distributions are log-transformed to prevent such features from excessively distorting clustering. As it is only present in one of the two hospital, EtCO2 is also dropped before analysis to ensure a representative comparison across hospitals. For interpretability, a random subset of 2500 points are plotted, and the results are presented in Figure 6.

In the two dimensional embedding, Figure 6 demonstrates high separability across hospitals. As the underlying data is unlikely to differ substantially across ICU units, the high separability is likely driven by differences in measurement recording practices, especially for sparse lab measurements. Although this difference may have been exaggerated by imputing a large number of missing points, this separability still presents a challenge for model generalisation across hospitals, especially if models over-fit to individual hospital patterns.

Notably, sepsis points are not clustered in the two dimensional embedding. Instead they are dispersed across clusters, likely reflecting that demographic variability and hospital measurement differences are more dominant separating factors than sepsis onset itself. This highlights the importance of patient trajectories for accurate early predictions, something which t-SNE cannot capture.

IV. Exploratory Modelling

In order to predict sepsis, a range of machine learning methods are employed and adapted to the context at hand. This involves exploration of different model types, preprocessing and postprocessing methods for early prediction.

A. Preprocessing

For all models the data is partitioned using a patient-level split, ensuring time-series data for each patient remains in a single subset. Patients are assigned to training, validation and test sets using a *60/20/20* split, stratified by the patient-level sepsis label. This addresses the high class imbalance, ensuring a representative number of sepsis patients are included in each subset.

B. Evaluation Protocol

All tuning decisions are made exclusively on the validation set. The test set is excluded and only evaluated on with the final tuned models. The decision threshold is selected by maximising the F1 score on the validation set. Under severe class imbalance, metrics like accuracy can be misleading — a model predicting non-sepsis for every time step would still achieve over 98% accuracy. F1, as the harmonic mean of precision and recall, requires the model to both detect septic cases and avoid false alarms, making it a more balanced measure of performance. It is also preferred over the PhysioNet utility score, which encodes timing assumptions specific to sepsis, making F1 more comparable across tasks. The utility score is also substantially slower to compute, making it impractical for iterative tuning.

C. Base Models

Logistic regression serves as one of our two baseline models. It passes a linear combination of input features through a sigmoid function, mapping predictions to a probability between 0 and 1, which is then thresholded to produce a binary label. Despite its simplicity, logistic regression remains a well-established approach for clinical prediction tasks; a

systematic review of ML models applied to sepsis mortality found it to be the most frequently used method across studies, appearing in 22 of 31 reviewed papers [12]. However, logistic regression assumes a linear decision boundary, and cannot model complex feature interactions, resulting in an F1 of 0.0773. We aim to quantify the performance gains offered by more complex models.

The Extreme Gradient Boosting (XGBoost) model served as our main baseline model, a decision-tree-based ensemble algorithm that operates within a gradient boosting framework [13]. XGBoost builds an ensemble of weak learners trained to correct the residual errors of the previous one, allowing it to capture complex non-linear relationships between features. It also handles missing values natively, which is particularly valuable for this dataset. XGBoost has demonstrated strong performance in sepsis prediction [14]. A key advantage of XGBoost over black-box models is its compatibility with SHAP (SHapley Additive Explanations), enabling interpretability of model predictions [15]. Its primary limitation for this dataset is the nature of its snapshot predictions, which can only account for measurements at the current time-step. As a result, temporal reasoning must be encoded manually rather than learned - later motivating the implementation of sequential model which are more naturally suited to this application.

D. Feature Engineering

1) Temporal Features: The base models treat each hourly observation as an independent snapshot, with no access to how a patient's state had been evolving. Adding temporal features, such as rolling-window statistics, helps capture these patterns. For each vital sign, the mean, standard deviation, minimum, and maximum are computed from the raw data over rolling windows of 3 and 6 hours. These rolling stats are not suited to sparse data, and so are not engineered for lab measurements or EtCO2.

The trend for each vital sign can also help capture temporal information. For this, the rates of change across 1, 3, and 6 hours intervals and the deviation from the 6-hour rolling mean are engineered for each vital sign.

2) Abnormal Values: Further features are added to mark abnormal clinical values, defined as being outside of normal ranges. Thresholds are selected using a clinically informed combination of standard adult reference ranges, historical SIRS criteria [16], and Sepsis-3 definitions for hypotension and lactate abnormalities [17]. This included: if a value is abnormal at the current hour, the cumulative count of abnormal hours so far for the patient, and the hours since the last abnormal reading.

3) Cross-Feature Interaction Ratios: The final features to be added are for cross-feature interaction ratios. Shock index (heart rate / systolic blood pressure) and narrow pulse pressure (systolic blood pressure – diastolic blood pressure) can signal shock, while elevated BUN-Creatine ratios suggest pre-renal kidney injury.

4) Missingness Features: Some clinical values are only taken when the doctor feels it is necessary, suggesting that they could be particularly relevant to sepsis prediction. Removing or imputing these features risks losing this implicit information, so engineering new features to capture missingness patterns may help preserve it [18]. Before the features are filled, missingness indicators are introduced to track which values are missing. For each clinical variable (7 vitals and 26 labs), a binary flag indicating if the value is missing, and the number of hours since the last measurement was recorded are added. Two additional aggregate counts are also added for the number of vitals and labs missing.

All features are computed using only past and current data for each patient to avoid future leakage. In total, 188 new features are added.

E. Imputation

Although LI is the most accurate imputation method (see section III-A), it cannot be used for time-series modelling. This is because it interpolates using future data points, creating data leakage that would potentially inform models using data that, in practice, would not be available at the time of prediction. As the second most accurate imputation technique, FF is an alternative that does not leak future information and is therefore the chosen imputation method. The removal of features with large amounts of data is also considered, especially for the sparse lab measurements. However, as these may be highly valuable to early sepsis detection, they are not removed until feature tuning when their importance to model predictions is known.

Given the high amounts of missing data, especially in lab features, filling of the whole dataset may distort patient trajectories and weaken model performance. To assess the impact of different levels of imputation, XGBoosting models are evaluated on: FF of all vital and lab features, FF of only vital signs, FF of only vital signs up to a three-hour limit, and no filling of vitals or labs. If no limit is placed on filling, median fill, calculated only on the training data, is used to fill any remaining missing values.

TABLE II
COMPARISON OF IMPUTATION METHODS FOR XGBOOST MODEL.

Imputation Method	AUROC	AUPRC	F1	Utility
FF vital + lab	0.8400	0.1071	0.1840	0.2122
FF vital only	0.8403	0.1113	0.1882	0.2311
FF vital only 3h limit	**0.8402**	**0.1105**	**0.1883**	**0.2323**
No imputation	0.8398	0.1104	0.1878	0.2336

From the results in Table II, it is found that the best level of imputation for the XGBoost models is forward filling only vital signs up to a three-hour limit. This provides the best balance between increasing the information available to model while limiting the distortion of patient trajectories introduced by the false stability of forward filling large amounts of data.

V. XGBOOST PIPELINE

After exploratory modelling, a full modelling pipeline for a temporal XGBoost is established.

A. Dataset and Experimental Setup

After applying the feature engineering pipeline and imputation, the final modelling dataset contained 232 features. Since only a small fraction of hourly observations correspond to a positive sepsis indicator, the model is vulnerable learning to predict a patient as non-septic at almost every time. To counteract this bias, XGBoost's scale_pos_weight is set to 54.26, weighting each positive sample roughly 54 times more heavily during training. The model is regularised primarily through a low learning rate (0.05) to help reduce overfitting.

B. Feature Selection

A baseline temporal model, trained on all 232 features, achieved a ROC-AUC of 0.8406, PR-AUC of 0.1080, and F1 of 0.1917 on the validation set (threshold 0.78). To determine whether a reduced subset could improve generalisation, two feature selection strategies are evaluated: progressively dropping features ranked by XGBoost's built-in importance (gain), and selecting features with positive permutation importance.

1) Built-in importance ranking: Models are trained at feature count thresholds ranging from 1 to 232 and evaluated on the validation set. As shown in Figure 7, F1 varied by less than 0.005 across all thresholds above 20 features, peaking at 200 features while dropping sharply below 20.

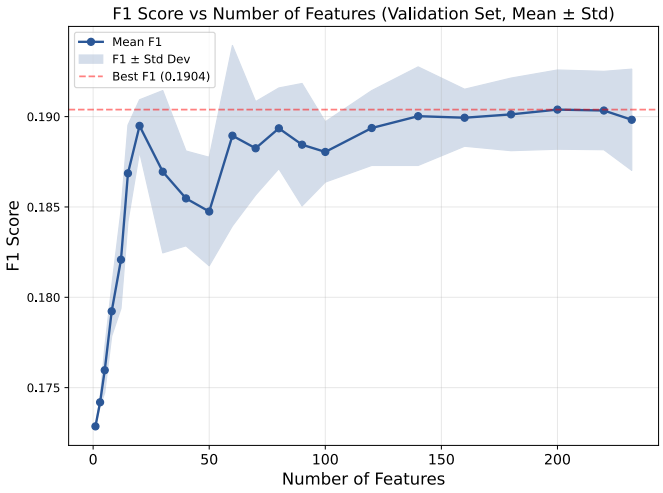


Fig. 7. Validation F1 metric as a function of the number of features selected by built-in importance

2) Permutation importance: As an alternative approach, each feature is randomly shuffled in turn, and the resulting change in validation performance (average precision) is measured over 5 repeats. This identified 149 features with positive importance and 83 with zero or negative importance, meaning that shuffling those 83 features did not degrade — and in some cases actually improved — predictions. Table III compares the three candidate feature sets.

The permutation-selected model achieved the strongest PR-AUC and F1, and is therefore selected as the final configuration.

Figure 8 illustrates the full XGBoost modelling pipeline.

TABLE III
COMPARISON OF FEATURE SELECTION STRATEGIES ON THE VAL SET.

Feature Set	ROC-AUC	PR-AUC	F1
Full (234)	0.8406	0.1080	0.1917
Top 200 (built-in)	0.8410	0.1085	0.1904
Perm. 149	**0.8410**	**0.1130**	**0.1918**

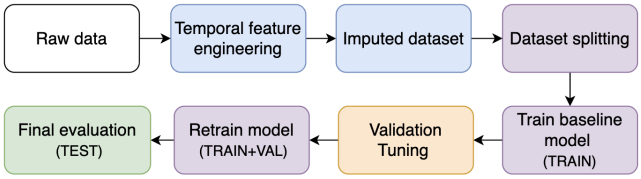


Fig. 8. Overview of the XGBoost modelling pipeline.

C. Final Model and Test Results

The final model used the 149 permutation-selected features with a decision threshold of 0.77, both chosen entirely on the validation set. It is retrained on the combined training and validation data with the threshold locked, and evaluated once on the held-out test set (8,068 patients).

TABLE IV
TEST SET PERFORMANCE OF THE FINAL XGBOOST MODEL VS. THE PHYSIONET BASELINE.

Model	AUROC	AUPRC	F1	Utility
PhysioNet Baseline	0.7001	0.0604	0.0772	0.1890
Base XGBoost	0.8115	0.1096	0.1816	0.2221
XGBoost + Temporal	**0.8455**	**0.1207**	**0.1977**	**0.2932**

The XGBoost model substantially outperformed the PhysioNet baseline across all metrics, with the utility score improving from 0.189 to 0.2932, confirming clinically meaningful early detection performance. In comparison, the highest scoring paper from the original challenge achieved a utility score of 0.36, slightly outperforming the temporal XGB model.

VI. SEQUENTIAL MODELS

As sepsis is a time-evolving condition, sequential models are well suited to sepsis prediction, as they can directly learn the temporal trajectories of patients without requiring manual feature engineering [19]. The Gated Recurrent Unit (GRU) is a recurrent neural network that processes sequential data one time step at a time, using its hidden state to maintain temporal context across the sequence. Sequential models have been shown to capture these patterns more effectively than static approaches for early sepsis detection [20]. The GRU performance is analysed against XGBoost to evaluate how inherent sequential reasoning compares to the engineered temporal features.

The GRU model is implemented using the Keras libraries. As GRU cannot natively handle any missing values, the whole dataset is imputed using forward fill followed by median fill for any remaining gaps. The dataset is split as described in Section IV-A. GRU is evaluated on two variants of the dataset:

one with missingness indicator features appended to the input before imputation, and one with both missingness and time-aware clinical features. No additional temporal features are added, as the GRU is designed to learn temporal structure directly from the sequence [19].

GRU requires training on data points of the same shape [21]. Therefore, the patients' sequences are padded to the global maximum patient length using a padding value of -1.0, with masking applied to ensure the model ignores padded time steps. Padding to the global maximum rather than truncating is chosen deliberately: a proportionate number of long-stay patients are septic, and given the class imbalance in the dataset, the memory overhead is considered an acceptable trade-off to avoid discarding any septic cases. Models are trained for up to 15 epochs with early stopping and a batch size of 64. The results for the GRU are presented in Table V.

TABLE V
GRU MODEL PERFORMANCE ON THE TEST SET ACROSS FEATURE CONFIGURATIONS.

Feature Set	AUROC	AUPRC	F1	Utility
No Features	0.7158	0.0703	0.1578	0.2591
Missingness	**0.7608**	**0.0840**	**0.1592**	**0.2274**
Miss.+Time-Aware	0.7649	0.0614	0.1259	0.1955

The GRU achieved an AUROC of 0.715 with raw features, improving to 0.764 with the addition of missingness and time-aware features. However, adding time-aware features did not consistently improve performance, as F1 and utility both declined. AUPRC remained low across all configurations (0.06–0.08), reflecting the severe class imbalance and indicating the model generates a high rate of false alarms despite weighted loss reweighting.

Building on the issues of missingness in sequential models, previous research has attempted to overcome this through the development of models that make use of informative missingness. Gated Recurrent Unit with Decay (GRU-D) is an example of this, and is a specialised variant of GRU developed specifically for medical contexts [22]. As such, the GRU-D is implemented using the PyPots toolkit [23], and the model is trained on full access to patient datasets, allowing it to realise common patient trajectories. Since GRU-D is designed to incorporate missing values, no imputation is required. When testing, to avoid time leakage and produce prediction results per hour, the model is given patient data up to the point of row i in order to make the prediction of sepsis likelihood at time i.

Comparative results of the sequential models are presented in Table VI. The GRU-D achieved a utility of 0.253, slightly below the GRU-s best utility of 0.259. Both sequential models slightly outperform the baseline model, but underperform when compared to the temporal XGBoost model.

This could be due to the fact that, for the both GRUs, every patient file had to be either padded or constrained to the same shape, abstracting some of the detail and nuance away from the ground truth. Additionally, by the nature of how the prediction is conducted (on subsets of patient data), the models are implemented outside of their intended utility. Finally, XGBoost is more robust to class imbalance and missing values by design, while the sequential models require larger labelled datasets to best perform [24] [25].

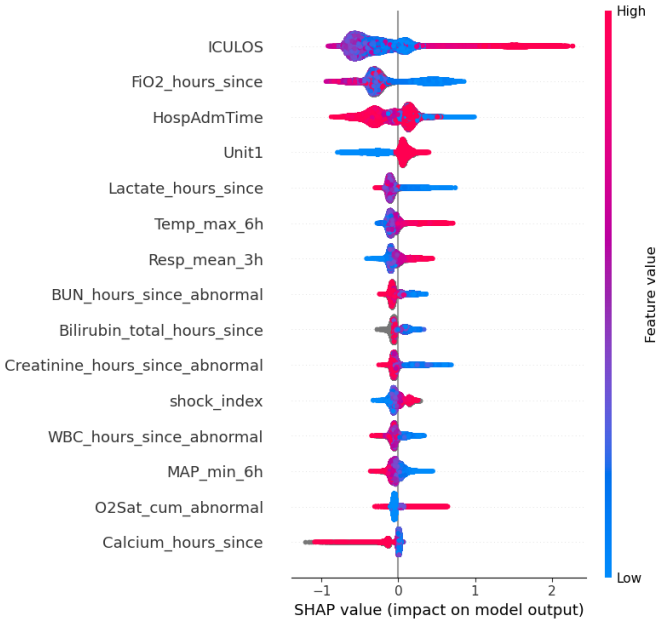


Fig. 9. Top 15 features by SHAP analysis for the XGBoost model.

TABLE VI
COMPARISON OF FINAL MODELS ON THE TEST SET.

Model	AUROC	AUPRC	F1	Utility
XGBoost	**0.8455**	**0.1207**	**0.1977**	**0.2932**
GRU	0.7168	0.0703	0.1578	0.2591
GRUD	0.6870	0.0632	0.1301	0.2309

VII. ANALYSIS

In order to gain a clearer understanding of how features contributed to Sepsis Prediction, Shapley plots are produced to illustrate in greater detail the patterns learned by the XGBoost model. SHAP plots encode importance to the model along the x-axis and feature value in colour, allowing for interpretation of how values led to sepsis prediction. In this analysis, exploration is split between medical and process features. Medical features are those directly related to patient health and serve to both confirm existing medical knowledge of sepsis indicators as well as understand how these features change in the 6 hours before Sepsis diagnosis. Process features are more indicative of the medical procedure for ICU patients. By examining both of these dimensions, this analysis aims to provide a comprehensive overview of septic data symptoms and management within ICU settings.

Three of the top features included context variables - ICULOS (ICU Length of Stay), Hospital Admission Time and whether a patient is in the Medical ICU (Unit1) as opposed to the Surgical ICU. ICULOS, representing the patient's length of stay in

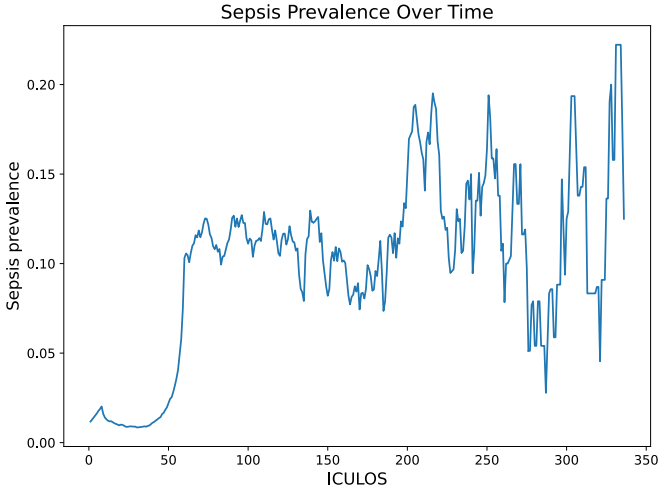


Fig. 10. Sepsis rate by ICULOS across both hospitals.

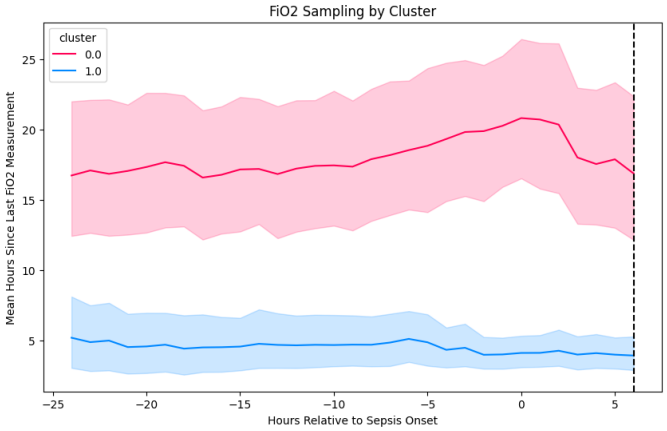


Fig. 11. FiO2 Sampling by Cluster

the ICU up to that point, ranks first by SHAP dependence with higher values pushing predictions towards sepsis. As visible in Figure 10, the proportion of septic patient increases over time. Consequently, the model may be learning a temporal bias, associating longer ICU stays with a higher likelihood of sepsis, particularly given most patient stays are under 60 hours. Hospital admission time is similarly informative to the models, as a patient who moved quickly from a general ward into the ICU is considered more worrisome, as well as patients in the Medical ICU, who are more likely to develop sepsis than those in the Surgical ICU.

A. *Medical Features*

In terms of medical features, the model tended to rely on rolling windows and temporal versions of features as opposed to their true baselines, indicating that trajectory is most important when predicting sepsis. The only true baseline feature to appear in the top 25 features by SHAP analysis is Temperature. As a known indicator, the model learns that high temperatures imply sepsis, as well as steep temperature trajectories. Interestingly, Temperature is also the leading factor in false positive predictions, likely because it is a symptom associated with many other illnesses. This indicates that temperature, although useful as a sepsis indicator, must be considered in conjunction with other features.

Abnormal levels of Creatinine are also considered sepsis indicators, with the model considering recent abnormal creatinine measurements concerning. This is in line with patients exhibiting symptoms of Sepsis-Induced Acute Kidney Inflammation (AKI). However, sepsis-related creatinine trajectories do not follow one pattern. This is in line with our findings since the second most likely feature to contribute to a false negative sepsis prediction is low creatinine, likely because the model primarily learns one pattern of creatinine trajectory. Similarly, another important feature linked to sepsis is BUN (Blood Urea Nitrogen), for which the model also looks at the time since abnormal.

This overall medical analysis confirms the model is utilising many of the same indicators of sepsis used in hospitals by clinicians and focuses primarily on trajectory.

B. *Process Features*

The Sepsis diagnosis process is one particularly reliant on clinician response and testing, as many septic symptoms are also present in other ICU illnesses, and there is no diagnostic test to confirm sepsis. Therefore, given the nature of medical data collection, the model learns this informative missingness as an important factor in sepsis diagnosis. Although a less strictly medical measure with which clinicians would be able to diagnose sepsis, it is useful to explore these metrics to understand how sepsis is monitored and diagnosed as a process.

In order to better understand patient trajectories, K-means clustering is performed on the SHAP values of positive patient predictions with the intent of categorising differing trajectories into sepsis. This results in two distinct clusters, separated by process features.

Exemplified by the FiO2 plot in Figure (11), it is clear that one cluster of patients is categorised by much more frequent testing than the other - a pattern repeated across many of the lab signals. This disparity indicates that within septic patients, there is a group presenting as more overtly septic to the clinicians, which could be patients the hospital is concerned about and therefore monitored more heavily. Opposing this, is another group of patients who are also eventually diagnosed with sepsis, but had testing performed much less frequently in the lead up to their diagnosis. This could be patients who fell under the radar, had less extreme pre-sepsis conditions, or exhibited septic symptoms in rapid deterioration.

This indicates that process measures are useful when categorising patient treatment in order to better examine trajectories in detail. These results also imply that patient trajectories are structured not only by medical measurements, but also by the pattern of data collection. This suggests that this ICU data reflects two simultaneous dimensions - the evolution of patient physiology and the pattern of intensity of clinical observation.

Interestingly, when examining base medical data such as Heart Rate or Blood Pressure in this cluster, patients subject to more intensive monitoring actually exhibited more stable trends. Initially, this appears counterintuitive as these patients are clinically considered to be at a higher risk; however, this could be explained by the inherent link between observation and treatment in ICU settings. Patients who are more frequently monitored are also more likely to receive aggressive interventions, which stabilise vital signs. As a result, the dataset is more representative of the effects of clinical management and is a potential reason why our model doesn't rely on what would usually be considered critical medical data in sepsis diagnosis. Stability in vital signs does not necessarily indicate a stable patient, but instead reflects a high level of clinical intervention. This reinforces that clinical datasets such as this one are influenced by observer effect and treatment biases, and are not representative of pure medical data. This highlights the importance of the interpretation of medical data within the context of measurement reasoning and timing. Additionally, much of this analysis emphasises the need for expert knowledge when interpreting data from highly specialised contexts such as medical data.

C. Cross-Hospital Generalisability

To evaluate generalisability across hospitals, the XGBoost model underwent retraining with the final feature set and hyperparameters, using one hospital for training and the other for external testing. From this, the cross-hospital performance is weak in both directions, with F1 scores of 0.124 and 0.129, shown in Table VII. Confusion matrices showed low sensitivity, with many positive septic time points being missed. However, sepsis onset timing is similar across hospitals, suggesting that timing differences are unlikely to be the cause of poor transfer.

To investigate whether the feature set encoded hospital-specific patterns, a separate hospital-classification model is trained to predict which hospital each hourly time-step originated from. This achieved an AUROC of 0.9986, showing that the final feature set almost perfectly encoded hospital identity. Several of the most hospital-predictive features are features reflecting hospital monitoring and recording practices, particularly measurement-timing and missingness features.

TABLE VII
CROSS-HOSPITAL PERFORMANCE BY FEATURE SET.

Features	Test	AUROC	AUPRC	F1	Utility
Original	A → B	0.7317	0.0617	**0.1241**	**0.1129**
Original	B → A	0.7471	0.0689	**0.1290**	**0.1321**
Removed Top 20	A → B	0.7889	0.0755	0.1393	0.1719
Removed Top 20	B → A	0.7521	0.0823	0.1513	0.1531
Clinical only	A → B	0.7610	0.0655	**0.1332**	**0.1878**
Clinical only	B → A	0.7542	0.0909	**0.1650**	**0.2240**

These hospital-specific features are also important to the sepsis model, suggesting that it partly relied on monitoring and workflow patterns rather than only transferable physiological deterioration signals. Prediction behaviour also differed; when trained on hospital B and tested on hospital A, the model produced higher sepsis probabilities and more positive alerts, suggesting that hospital A's feature distribution pushed predictions more strongly towards sepsis and contributed to poorer calibration across hospitals. Removing monitoring and recording features and only using clinical features slightly improved the F1 score and the utility (see Table VII), suggesting that poor external generalisation is partly driven by hospital-specific measurement and workflow patterns.

D. Demographic Features

Patient-level outcomes are broadly similar across age and gender groups, with no strong evidence of substantial age or gender related bias. Small differences are observed across age bands, with false positive proportions somewhat higher in the 20-29, 70-79 and 80-89 groups. For gender, male patients have slightly higher false positive and false negative proportions, as well as a higher observed sepsis rate. However, these differences are small and may be influenced by subgroup size differences and variation in underlying sepsis prevalence, so they do not provide clear evidence of bias.

VIII. Limitations and Future Work

A significant limitation of the implemented models is the poor generalisation across hospitals (Section VII-C), which could be due to several factors. For example, hospitals differ in their clinical practices and workflows, which may affect the timing of measurements, their accuracy, and whether certain measurements are taken at all. This is consistent with previous work on the PhysioNet datasets, where algorithms overfit on known databases and underperform on hidden datasets [7]. To improve the generalisability, more data from a wider range of hospitals would be needed for model training.

Another limitation is the potential bias introduced by differing lengths of stay for patients in the ICU. For each hour that a patient is in the hospital, a row of data is added to the dataset. This means that patients who stay longer contribute more data points. This is especially noticeable for septic patients, as they are only kept in the dataset for six hours after they are diagnosed. In the future, the models could investigate strategies to either weight or remove excessive time-series data from individual patients.

The time-series data also includes a large quantity of missing data, meaning the model predictions are sensitive to the preprocessing implemented as well as the data itself. This makes it hard to have sufficient confidence in the results to be used in medical practice.

Medical professionals may also find it more useful for the predictions to be made across different time frames, rather than just looking at the previous six hours. This could be tackled in the future by investigating if predicting earlier or later impacts the accuracy. Furthermore, the implemented models assume a uniformly sampled time series data at hourly intervals, which may not reflect the measurement practices of all hospitals. In practice, clinical measurements are often recorded irregularly, making the ability to handle non-uniform time series a critical requirement for real-world deployment.

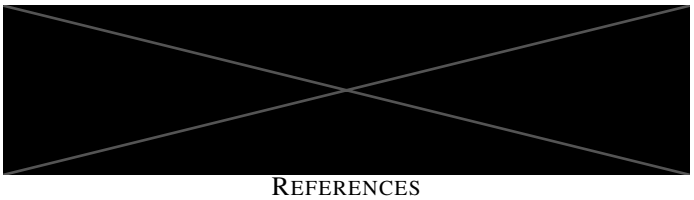
IX. Conclusion

During the course of this study, exploration was conducted into how sepsis data varies across many dimensions, including age, gender and hospitals. Multiple models - Logistic Regression, XGBoost and GRU - were implemented and evaluated within the context of early sepsis prediction. Through comprehensive pre and post processing exploration, model performance was improved and the final model was able to identify trajectory features and abnormal values as key for the early prediction of sepsis. This demonstrates the utility of designing features tailored to the context using background knowledge, especially in highly specialised situations such as sepsis.

The investigation also highlights the importance of contextual factors - particularly the effect of treatment on recorded medical data. These findings could help improve the medical data collection process and emphasise the importance of context and expert knowledge when interpreting complex datasets. Additionally, sepsis management processes differ by hospital, affecting any predictions based on hospital measurements. These findings could be used by medical professionals from hospitals A and B to target key measurements in data collection. This would help them reduce redundant tests, saving time and resources. To improve the generalisability, more data across a wider range of communities would be needed, as well as further contextual data such as illnesses pre-sepsis. By widening data access, interpretation, and prediction may improve, but considerations regarding privacy and ethics must be made.

Code

URL: https://github.com/Ava-rsn/sepsis_prediction/
Our AI Declaration can be found in the GitHub linked above.



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